

Problem A. Concentration inequality

In this problem, we will derive a concentration inequality that is finer than Hoeffding's inequality.

Let $X_1, \dots, X_m$ be independent random variables drawn according to some distribution $\mathcal{D}$ with mean p and support included in $[0, 1]$. Let $q \in [0, 1]$ and define $\hat{p}$ by $\hat{p} = \frac{1}{m} \sum_{i=1}^m X_i$.

1. Use Markov's inequality to prove that for any $t > 0$,

$$\mathbb{P}[\hat{p} \geq q] \leq e^{-tmq} \prod_{i=1}^m \mathbb{E}[e^{tX_i}].$$

2. Show that

$$\mathbb{P}[\hat{p} \geq q] \leq [e^{-tq} (1 - p + e^tp)]^m$$

(Hint: you can show that $\forall x \in [0, 1], e^{tx} \leq 1 - x + e^tx$).

3. Choose $t > 0$ to minimize the right-hand side and prove the following inequality:

$$\mathbb{P}[\hat{p} \geq q] \leq e^{-mD(q||p)},$$

where $D(q||p) = q \log \frac{q}{p} + (1 - q) \log \frac{1-q}{1-p}$ is the binary relative entropy of p and q .

4. Show that this can be used to derive a bound that is finer than Hoeffding's inequality by choosing $q = p + \varepsilon$, for any $\varepsilon > 0$ (Hint: you can use Pinsker's inequality, which in this context gives $D(q||p) \geq 2|p - q|^2$).

(1) Recall that if Y is a nonnegative random variable and $a > 0$, then Markov's inequality states:

$$\mathbb{P}[Y \geq a] \leq \frac{\mathbb{E}[Y]}{a}.$$

Here, let

$$Y = e^{tm\hat{p}} \quad \text{and} \quad a = e^{tmq}.$$

Then,

$$\mathbb{P}(e^{tm\hat{p}} \geq e^{tmq}) \leq \frac{\mathbb{E}[e^{tm\hat{p}}]}{e^{tmq}}.$$

Since the event $\{e^{tm\hat{p}} \geq e^{tmq}\}$ is equivalent to $\{\hat{p} \geq q\}$, we have:

$$\mathbb{P}(\hat{p} \geq q) \leq e^{-tmq} \mathbb{E}[e^{tm\hat{p}}].$$

Notice that

$$\hat{p} = \frac{1}{m} \sum_{i=1}^m X_i \implies e^{tm\hat{p}} = e^{t \sum_{i=1}^m X_i}.$$

Since the X_i are independent, the expectation of the exponential of the sum factors as a product:

$$\mathbb{E} \left[e^{t \sum_{i=1}^m X_i} \right] = \prod_{i=1}^m \mathbb{E} \left[e^{t X_i} \right].$$

Thus, we obtain:

$$\mathbb{P}(\hat{p} \geq q) \leq e^{-tmq} \prod_{i=1}^m \mathbb{E} \left[e^{t X_i} \right].$$

(2) Define the function

$$f(x) = e^{tx}.$$

We compute its second derivative:

$$f''(x) = t^2 e^{tx}.$$

Since $t > 0$ and $e^{tx} > 0$ for all x , we have $f''(x) > 0$ for all x . This shows that $f(x)$ is a *convex* function.

For any $x \in [0, 1]$, we can express x as a convex combination of 0 and 1:

$$x = (1 - x) \cdot 0 + x \cdot 1.$$

Using the property of convex functions, we have:

$$f((1 - x) \cdot 0 + x \cdot 1) \leq (1 - x)f(0) + xf(1).$$

the inequality becomes

$$e^{tx} \leq (1 - x) \cdot 1 + x \cdot e^t = 1 - x + e^t x.$$

This completes the proof of hint.

Taking expectation with respect to X_i yields

$$\mathbb{E} \left[e^{t X_i} \right] \leq \mathbb{E} \left[1 - X_i + e^t X_i \right].$$

Because expectation is linear, we have

$$\mathbb{E} \left[1 - X_i + e^t X_i \right] = 1 - \mathbb{E}[X_i] + e^t \mathbb{E}[X_i] = 1 - p + e^t p.$$

Since the same bound holds for each X_i , we get

$$\prod_{i=1}^m \mathbb{E} \left[e^{t X_i} \right] \leq (1 - p + e^t p)^m.$$

Substituting this into our earlier bound gives

$$\mathbb{P}(\hat{p} \geq q) \leq e^{-tmq} (1 - p + e^t p)^m.$$

Since $e^{-tmq} = \left(e^{-tq}\right)^m$, we can write:

$$\mathbb{P}[\hat{p} \geq q] \leq \left[e^{-tq}(1 - p + e^t p)\right]^m.$$

This completes the proof.

(3)

Let

$$f(t) = e^{-tq}(1 - p + p e^t).$$

We wish to choose $t > 0$ so that $f(t)$ is minimized.

Define

$$\phi(t) = \ln f(t) = -tq + \ln(1 - p + p e^t).$$

Differentiating with respect to t gives

$$\phi'(t) = -q + \frac{p e^t}{1 - p + p e^t}.$$

Setting $\phi'(t) = 0$ yields

$$\frac{p e^t}{1 - p + p e^t} = q.$$

Solving for e^t :

$$p e^t = q(1 - p + p e^t) \implies p e^t = q(1 - p) + q p e^t.$$

Rearrange the terms to isolate e^t :

$$p e^t(1 - q) = q(1 - p) \implies e^t = \frac{q(1 - p)}{p(1 - q)}.$$

Thus, the minimizing value is

$$t^* = \ln\left(\frac{q(1 - p)}{p(1 - q)}\right).$$

Substitute $t = t^*$ into the bound:

$$f(t^*) = e^{-qt^*}(1 - p + p e^{t^*}).$$

Since

$$e^{t^*} = \frac{q(1 - p)}{p(1 - q)},$$

we have:

$$e^{-qt^*} = \left(\frac{q(1 - p)}{p(1 - q)}\right)^{-q},$$

and

$$1 - p + p e^{t^*} = 1 - p + p \frac{q(1 - p)}{p(1 - q)} = 1 - p + \frac{q(1 - p)}{1 - q} = (1 - p) \left(1 + \frac{q}{1 - q}\right) = \frac{1 - p}{1 - q}.$$

Thus,

$$f(t^*) = \left(\frac{q(1-p)}{p(1-q)} \right)^{-q} \frac{1-p}{1-q}.$$

Rewrite the expression and we obtain

$$f(t^*) = \left(\frac{p}{q} \right)^q (1-p)^{1-q} (1-q)^{-(1-q)} = \left(\frac{p}{q} \right)^q \left(\frac{1-p}{1-q} \right)^{1-q}.$$

Observing that

$$e^{-D(q||p)} = \exp \left[-q \log \frac{q}{p} - (1-q) \log \frac{1-q}{1-p} \right] = \left(\frac{p}{q} \right)^q \left(\frac{1-p}{1-q} \right)^{1-q},$$

we conclude that

$$f(t^*) = e^{-D(q||p)}.$$

Thus, the optimal bound becomes

$$\mathbb{P}[\hat{p} \geq q] \leq [f(t^*)]^m = e^{-m D(q||p)}.$$

(4) choose

$$q = p + \varepsilon \quad \text{for any } \varepsilon > 0.$$

Then the bound becomes

$$\mathbb{P}[\hat{p} \geq p + \varepsilon] \leq \exp(-m D(p + \varepsilon || p)).$$

Pinsker's inequality, tells us that in this context

$$D(p + \varepsilon || p) \geq 2\varepsilon^2.$$

Substituting this into our bound, we obtain

$$\mathbb{P}[\hat{p} \geq p + \varepsilon] \leq e^{-2m\varepsilon^2}.$$

This bound is at least as strong (and under many circumstances finer) than Hoeffding's inequality. because:

- The binary relative entropy $D(p + \varepsilon || p)$ depends on the specific values of p and $p + \varepsilon$, thereby capturing the true large-deviation rate. In contrast, Hoeffding's inequality provides a uniform *worst-case* bound over all distributions with support in $[0, 1]$.
- Pinsker's inequality gives the lower bound $D(p + \varepsilon || p) \geq 2\varepsilon^2$, but $D(p + \varepsilon || p)$ is typically *greater* than $2\varepsilon^2$ unless $p = 1/2$ (or in the limit as $\varepsilon \rightarrow 0$). For example, when p is near 0 or 1, the divergence $D(p + \varepsilon || p)$ can be significantly larger than $2\varepsilon^2$, which implies a much faster exponential decay for the tail probability.

Problem B. PAC-Learning

1. **Concentric spheres.** Let $\mathcal{X} = \mathbb{R}^3$ and consider the set of concepts of the form

$$c = \{(x, y, z) : x^2 + y^2 + z^2 \leq r^2\} \quad \text{for some } r > 0.$$

Show that this class can be (ε, δ) -PAC-learned from training data of size

$$m \geq \frac{1}{\varepsilon} \log\left(\frac{1}{\delta}\right).$$

You should carefully justify all steps of your proof. (For this problem, you should seek a proof similar to one presented in class for rectangles in the plane and not leverage a more powerful or more general theorem.)

Problem C. Rademacher complexity

1. Let $\mathcal{H}_1$ and $\mathcal{H}_2$ be two families of functions mapping $\mathcal{X}$ to $[0, 1]$ and let S be a sample of size m .

- (a) Show that for any sample S ,

$$\widehat{\mathfrak{R}}_S(\mathcal{H}_1 + \mathcal{H}_2) = \widehat{\mathfrak{R}}_S(\mathcal{H}_1) + \widehat{\mathfrak{R}}_S(\mathcal{H}_2).$$

- (b) Show that the empirical Rademacher complexity of $\mathcal{H}$ for any sample S of size m can be bounded as follows:

$$\widehat{\mathfrak{R}}_S(\mathcal{H}_1 \mathcal{H}_2) \leq \frac{3}{2} \left[\widehat{\mathfrak{R}}_S(\mathcal{H}_1) + \widehat{\mathfrak{R}}_S(\mathcal{H}_2) \right].$$

Hint: use the identity $x_1 x_2 = \frac{1}{4} [(x_1 + x_2)^2 - (x_1 - x_2)^2]$ valid for all $x_1, x_2 \in \mathbb{R}$ and Talagrand's contraction lemma.

2. Let $\mathcal{E}$ be the family of functions defined over $\mathcal{X}$ by $\mathcal{E} = \{x \mapsto \mathbb{1}_{x=z} : z \in \mathcal{X}\}$.

- (a) Let S be a sample of size m with distinct points. Prove the inequality $\widehat{\mathfrak{R}}_S(\mathcal{E}) \leq \frac{1}{m}$.
 (b) Let S be a sample of size m with a single point repeated m times. Prove the equality

$$\widehat{\mathfrak{R}}_S(\mathcal{E}) = \frac{1}{2m} \mathbb{E}_{\sigma} \left[\left| \sum_{i=1}^m \sigma_i \right| \right].$$

Show that $\widehat{\mathfrak{R}}_S(\mathcal{E}) \leq \frac{1}{2\sqrt{m}}$.

- (c) 1[bonus question] Let n^* denote the largest number of repetitions of any point in the sample. Give an upper bound on $\widehat{\mathfrak{R}}_S(\mathcal{E})$ in terms of n^* .